

Secure Virtualisation with SELinux

There's been much brouhaha over virtualisation. Have you ever wondered what would happen if a single compromised VM compromised all the other VMs running on the same hardware stack?

As the usage of virtualisation increases, the security that isolated physical machines enjoy is at risk. Rogue processes in a guest OS might break out of the virtual machine and get access to the host memory or other guests. Designing security features within hypervisors can control this. But in case of such incidents, the risks can be mitigated by integrating Mandatory Access Controls (MAC) with virtualisation.

With the integration of MAC-capable security schemes like SELinux, with virtualisation, the security position of guest machines can be improved.

Let's take a look at how the scenario might unfold. We can begin by examining the situation as it existed in the good old days when one physical machine ran a dedicated server:

- Before virtualisation, servers were generally physically isolated from each other (Figure 1). An attacker taking over one server would typically have control over just that. So, if he wanted to breach the other machines on the network, he essentially had to launch a network-based attack to exploit them. This limited the damage of network attacks. To avoid these kinds of attacks, administrators used different tools like network firewalls, intrusion detection/prevention systems, and unified threat management systems to mitigate the risks.
- With the advent of virtualisation, several guest machines run on the same single host server (Figure 2), and probably as the same UID. This comes with its own risks. If an attacker gains access to a single virtual